ALGEBRAIZATION OF A COMPATIBLE SYSTEM OF MORPHISMS BETWEEN FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 10261-10288, immediately following the statement at 10238-10259.

Editorial note. The author ends with 'We omit the proof of uniqueness.'; that sentence is retained verbatim and the existence proof is complete inside the span.

Editorial note. Three cross-chapter pointers (Morphisms of Spaces) plus three same-chapter statements are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Algebraization of a formal closed subspace. chapter lemma lemma-algebraize-formal-closed-subscheme, cited in the proof. Produces the closed subspace Z inside X whose thickenings realise the given graphs.
- Passing a quasi-finite morphism through a finite one. chapter lemma lemma-quasi-finite-separated-pass-through-finite, cited at proof lines 10280-10281. Combined with properness of Z over X, converts the quasi-finite p into a finite morphism (Zariski's main theorem).
- Grothendieck's existence theorem. chapter theorem theorem-grothendieck-existence, cited at proof lines 10283-10284. Shows p_*O_Z = O_X because this holds modulo I^n for all n, whence p is an isomorphism.
- The quasi-finite part of a locally finite type morphism. Morphisms of Spaces, lemma spaces-morphisms-lemma-locally-finite-type-quasi-finite-part, cited at proof lines 10275-10276. Provides the open neighbourhood where p is quasi-finite, which properness then spreads to all of Z.

Source reference. Original lemma 08BG: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10146–10146. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 082K: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7755–7755. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NW: focused-seeds/geometry/S9-spaces-algebraize-morphism-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KP: focused-seeds/geometry/S9-spaces-algebraize-morphism-spaces/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted

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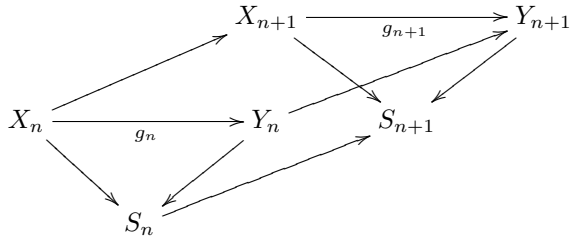
proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04NX: focused-seeds/geometry/S9-spaces-algebraize-morphism-spaces/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 08BE: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 9990–9990. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let A be a Noetherian ring complete with respect to an ideal I. Write $S = \mathrm{Spec}(A)$ and $S_n = \mathrm{Spec}(A/I^n)$. Let X, Y be algebraic spaces over S. For $n \geq 1$ we set $X_n = X \times_S S_n$ and $Y_n = Y \times_S S_n$. Suppose given a compatible system of commutative diagrams*



Assume that

(1) *$X \to S$ is proper, and*
(2) *$Y \to S$ is separated of finite type.*

Then there exists a unique morphism of algebraic spaces $g : X \to Y$ over S such that g_n is the base change of g to S_n.

Proof. The morphisms $(1, g_n) : X_n \to X_n \times_S Y_n$ are closed immersions because $Y_n \to S_n$ is separated (Morphisms of Spaces, Lemma 03KP). Thus by Lemma 08BG there exists a closed subspace $Z \subset X \times_S Y$ proper over S whose base change to S_n recovers $X_n \subset X_n \times_S Y_n$. The first projection $p : Z \to X$ is a proper morphism (as Z is proper over S, see Morphisms of Spaces, Lemma 04NX) whose base change to S_n is an isomorphism for all n. In particular, $p : Z \to X$ is quasi-finite on an open subspace of Z containing every point of Z_0 for example by Morphisms of Spaces, Lemma 04NW. As Z is proper over S this open neighbourhood is all of Z. We conclude that $p : Z \to X$ is finite by Zariski's main theorem (for example apply Lemma 082K and use properness of Z over X to see that the immersion is a closed immersion). Applying the equivalence of Theorem 08BE we see that $p_*\mathcal{O}_Z = \mathcal{O}_X$ as this is true modulo I^n for all n. Hence p is an isomorphism and we obtain the morphism g as the composition $X \cong Z \to Y$. We omit the proof of uniqueness. □